

ERMA SWARTZ

Urban Planner · Data & Spatial Analyst · Designer

New York, NY · 516.724.7700 · es4284@columbia.edu · ermaswartz.com · linkedin.com/in/erma-swartz · github.com/ErmaSwartz

Urban planner (Columbia GSAPP, May 2026) who builds the tools her analysis depends on — predictive models, spatial pipelines, scrollytelling sites, and a 3D-printed wearable air-quality sensor. Combines GIS and quantitative methods with policy fluency (CEQR, SEQRA, NEPA) to translate complex urban data into actionable planning recommendations. Thesis modeled 57M tCO₂e in potential emissions reductions for NYC construction.

SELECTED PROJECTS

Circular Economy & Embodied Carbon in NYC Construction — *Master's Thesis · Columbia GSAPP* 2025–2026

- Modeled embodied carbon across NYC building stock using OLS regression and Random Forest on DOB permits, EPDs, and material specs; identified floor area and structural system as dominant predictors of lifecycle emissions.
- Projected cumulative carbon savings under four policy pathways (procurement reform, subsidies, regulation, hybrid); hybrid scenario yields **57M tCO₂e** over 20 years and reaches market viability in 5 years.
- Conducted qualitative interviews with NYC construction practitioners and policymakers; synthesized quantitative results, regulatory analysis (CLCPA, EO 22/23, Buy Clean), and stakeholder input into policy recommendations.

Transit Access & Equity in New Jersey — *Independent Research · Built for NJTPA-style audience* 2025–2026

- Built tract-level ridership prediction model across 2,181 NJ census tracts using OLS and Random Forest; engineered spatial features from 31,000+ bus stops and 165 rail stations.
- Bus-stop density alone accounted for 55% of Random Forest predictive power — more than all demographic variables combined — reframing access (not demographics) as the primary driver of ridership.
- Delivered findings as a multi-page interactive site with maps, methods, and scenario tools; publicly viewable at ermaswartz.com/map.

NYC Congestion Pricing: Spatial Spillover Analysis — *Advanced Spatial Analysis · Columbia* Spring 2026

- Used difference-in-differences with a placebo year and Moran's I / LISA spatial statistics to trace how Manhattan CBD congestion pricing displaced traffic to surrounding bridges, outer-borough streets, and EJ communities.
- Combined MTA, traffic count, and satellite NO₂ data; published as a scrollytelling site using PySAL, Leaflet, and Chart.js.

Wearable Air-Quality Sensor ("Sensing Lungs") — *Design & Fabrication · Columbia* Spring 2025

- Designed and built a wearable environmental sensor from scratch — silicone wristband + lung-shaped 3D-printed housing in Rhino — integrating ozone, heart rate, GPS, and an ESP32 microcontroller.
- Collected paired biometric and environmental data across NYC; OLS and polynomial regression identified statistically significant cardiac response to ozone exposure at street level after 15+ minutes.

Citi Bike Equity-Weighted Expansion Optimization — *Urban Tech · Columbia* Spring 2025

- Analyzed 87MB of trip data to model demand, then built an optimization framework recommending 20 new station locations weighted by access gaps; equity weighting shifted 8 of 20 sites into outer-borough transit deserts a pure-ridership model would have ignored.

EXPERIENCE

Departmental Research Assistant · Columbia University GSAPP · New York, NY May 2025 – Present

- Cleaned, validated, and integrated multi-source spatial and tabular datasets (Census/ACS, DSNY, NYC DOB, field-sensor data) for ongoing urban-equity research at the census-tract level.
- Built reproducible Python workflows (pandas, GeoPandas) for spatial joins, geoprocessing, and tract-level cluster analysis; produced maps, technical memoranda, and visualizations supporting research outputs.
- Modeled 3D sensor housings in Rhino and supported deployment of a team-developed environmental sensor prototype, linking field collection to spatial analysis.

- Data Consultant & Application Developer** · Mandate Media · *Portland, OR (Remote)*

Jan 2024 – Jan 2025
- Built an internal React + AWS Amplify application with Python ETL backend that ingested client CSVs, matched donor records, and computed ROI / fundraising metrics — replacing a manual reporting workflow used across multiple congressional client accounts.
 - Automated recurring reporting in Python, cutting per-client analyst time and standardizing QA/QC across deliverables; designed dashboards and visualizations for technical and non-technical stakeholders.

- Policy & Data Analyst** · Retail Energy Advancement League · *Denver, CO*

Jan 2023 – Oct 2023
- Conducted quantitative analysis of energy markets, regulatory frameworks, and emissions data; authored technical memoranda and briefing materials advising leadership on long-term decarbonization strategy.
 - Translated regulatory research and stakeholder input into structured policy recommendations and legislative reports for external audiences.

Earlier Experience

- **Digital Account Manager**, 4Degrees, Denver, CO (2021–2022) — Led digital and email campaigns for Secretaries of State, Senate, and Congressional clients; produced content for one of the top-performing Democratic fundraising programs of the 2022 cycle.
- **Field Director**, Andrew Romanoff for Colorado, Denver, CO (2019–2020) — Managed field operations across 45+ counties; recruited and trained 10 staff; ran statewide Spanish-language voter outreach.
- **Field Manager**, Grassroots Campaigns Inc., New York, NY (2019) — Supervised field personnel; raised \$17,000 for Planned Parenthood in one summer.

EDUCATION

- Columbia University** · *M.S. Urban Planning, GSAPP*

Expected May 2026
- Coursework: Advanced Spatial Analysis · Urban Analytics · Planning Methods · GIS · Sensor Programming · Land Use & Zoning · Environmental Review (CEQR/SEQRA/NEPA)

Junior Editor, Urban Magazine (2024–2025)
- UC Berkeley Extension** · *Certificate, Data Analysis & Visualization*

May 2021
- University of Colorado Boulder** · *B.A. International Affairs*

Aug 2018
- Concentration in Latin American political economy and climate justice · Dean's List

TOOLKIT

- Data & Modeling**

Python (pandas, NumPy, scikit-learn, statsmodels) · Random Forest · XGBoost · OLS · scenario modeling · Jupyter
- GIS & Spatial**

ArcGIS Pro · GeoPandas · PySAL · Folium · Plotly · Moran's I / LISA · cartography
- Engineering & Cloud**

Git/GitHub · Google Cloud (gcsfs, large-scale geoJSON pipelines) · AWS Amplify · React · JSX · HTML/CSS
- Design & Fabrication**

Adobe Creative Suite (Illustrator, InDesign, Photoshop) · Rhino 3D · 3D printing · ESP32 sensor prototyping
- Policy & Communication**

CEQR · SEQRA · NEPA · technical memoranda · qualitative interviews · stakeholder briefings · Spanish (advanced conversational)